Applied field theory – fields, lines and waveguides

Circuit theory assumes that everything is small compared with the wavelength. When that no longer holds – in long cables, in antennas, in microwave circuits and around a high-voltage insulator – one must calculate with the fields themselves. “Applied field theory” was the course in which Maxwell’s equations became an engineering tool. The page has an interactive Laplace solver and a transmission line with a Smith chart.

> DTH, course 4820 (1987), 6 points: Applied field theory. The course built on the electromagnetism I had studied at Sussex, but with the engineer’s focus on solving concrete boundary-value problems. The transmission line and the waveguide have exactly the same mathematics as sound in pipes and ducts – a connection that runs through the thesis and the essay on category and congruence errors.

Vector analysis

The language of field theory is vector analysis. The gradient $\nabla\phi$ points in the direction in which a scalar field increases fastest; the divergence $\nabla\cdot\vec{D}$ measures how much a vector field “wells out” of a point; and the curl $\nabla\times\vec{H}$ measures how much it swirls. Two integral theorems tie the local and the global together:

$$\oint_S \vec{D}\cdot d\vec{A} = \int_V \nabla\cdot\vec{D}\,dV \quad\text{(Gauss)}, \qquad \oint_C \vec{H}\cdot d\vec{l} = \int_S (\nabla\times\vec{H})\cdot d\vec{A} \quad\text{(Stokes)}$$

Maxwell’s equations

In 1865 James Clerk Maxwell united electricity and magnetism in a set of equations, which Oliver Heaviside around 1884 wrote in the vector form we use today:

$$\nabla\cdot\vec{D}=\rho, \qquad \nabla\cdot\vec{B}=0, \qquad \nabla\times\vec{E}=-\frac{\partial\vec{B}}{\partial t}, \qquad \nabla\times\vec{H}=\vec{J}+\frac{\partial\vec{D}}{\partial t}$$

with $\vec{D}=\varepsilon\vec{E}$, $\vec{B}=\mu\vec{H}$ and $\vec{J}=\sigma\vec{E}$ in simple materials. The displacement current $\partial\vec{D}/\partial t$ was Maxwell’s own contribution; from it follow the wave equation and the speed of light $c=1/\sqrt{\mu_0\varepsilon_0}$. At the boundary between two materials the tangential component of $\vec{E}$ and the normal component

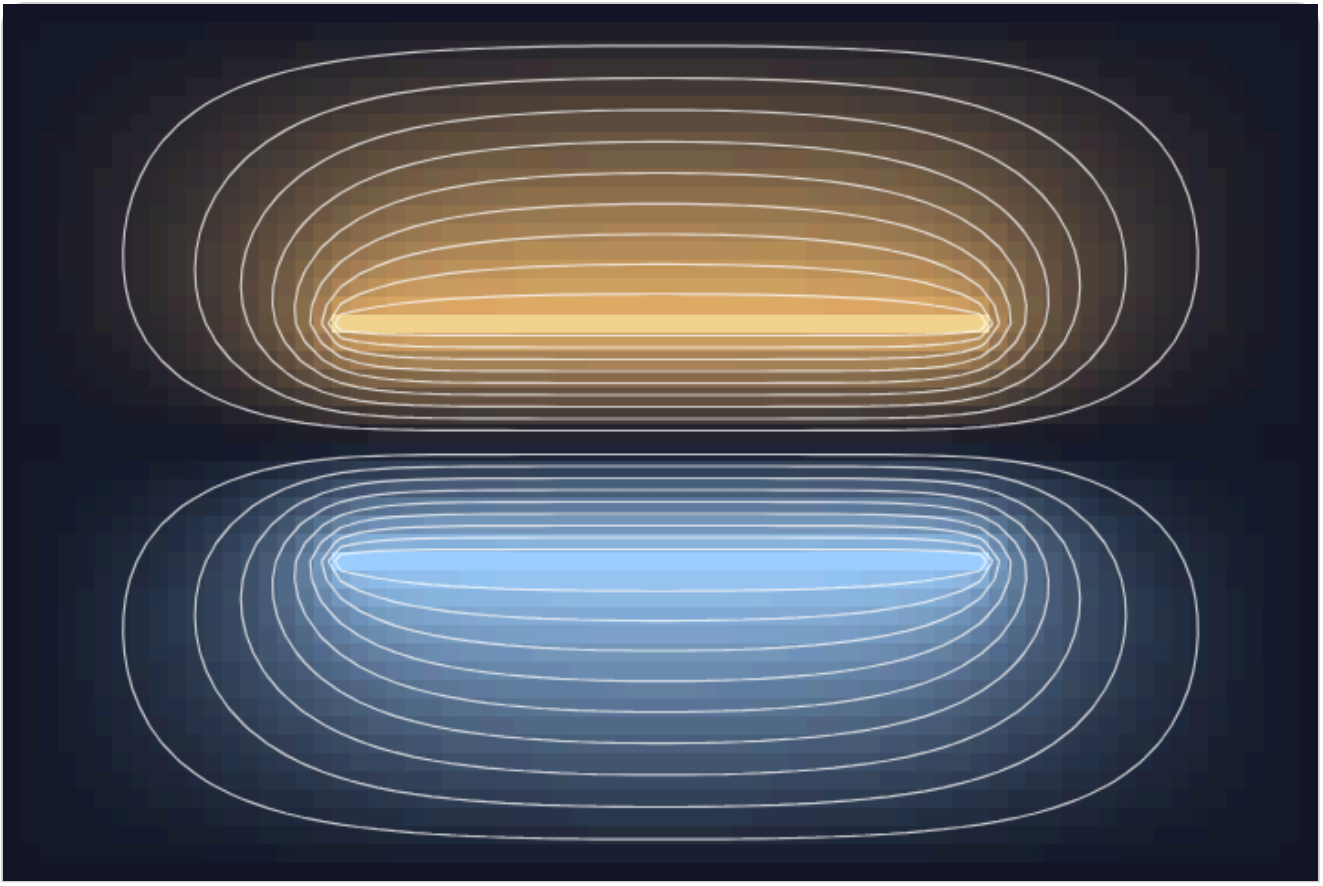
of $\vec{B}$ are continuous; on a perfect conductor the tangential $\vec{E}$ is zero. It is the boundary conditions that make field theory an engineering discipline. (History and physics: see electromagnetism.)

Laplace and boundary values

In a charge-free region the electrostatic potential is a solution of Laplace's equation $\nabla^2\phi = 0$. The solution is uniquely determined by the potential on the boundary – a *boundary-value problem*. Classical methods are separation of variables (with Fourier series in a rectangle and Bessel functions in a cylinder), the method of images (a charge above a conducting plane is equivalent to the charge and its mirror image) and conformal mapping, where in two dimensions complex functions map a difficult geometry onto an easy one. Laplace functions have a remarkable property: the value at each point is the mean of the values on a circle around the point. So there can be no maximum or minimum inside the region – which is why a charged particle cannot be trapped by static electric fields alone (Earnshaw's theorem, 1842).

Numerical solution

The mean-value property gives a simple numerical method: lay a grid over the region, fix the boundary values, and replace each interior point again and again by the mean of its four neighbours. This is Jacobi or Gauss–Seidel *relaxation*; Lewis Fry Richardson used it as early as 1910 with human computers. Successive over-relaxation (SOR) speeds up convergence by going a little beyond the mean. It is the forerunner of the finite difference and finite element methods that today solve field problems in three dimensions.



Skin effect

In a good conductor the displacement current is negligible, and Maxwell's equations become a diffusion equation. An alternating field penetrates the conductor only to a depth of the order of the skin depth

$$\delta = \sqrt{\frac{2}{\omega \mu \sigma}}$$

For copper δ is 9 mm at 50 Hz, 0.2 mm at 100 kHz and 2 µm at 1 GHz. At high frequencies the current flows only in the surface, and the resistance rises. That is why Litz wire is used in coils, silver plating on microwave components and hollow conductors in high-voltage substations. The same physics decides how deep an eddy current reaches into a workpiece in non-destructive testing.

Transmission lines

A long line – a coaxial cable, a circuit-board track, a telephone line – can be described by inductance L' and capacitance C' per metre (and losses R' and G'). Heaviside derived the *telegrapher's equations* in 1876–1887:

$$\frac{\partial u}{\partial x} = -L'\frac{\partial i}{\partial t} - R' i, \qquad \frac{\partial i}{\partial x} = -C'\frac{\partial u}{\partial t} - G' u$$

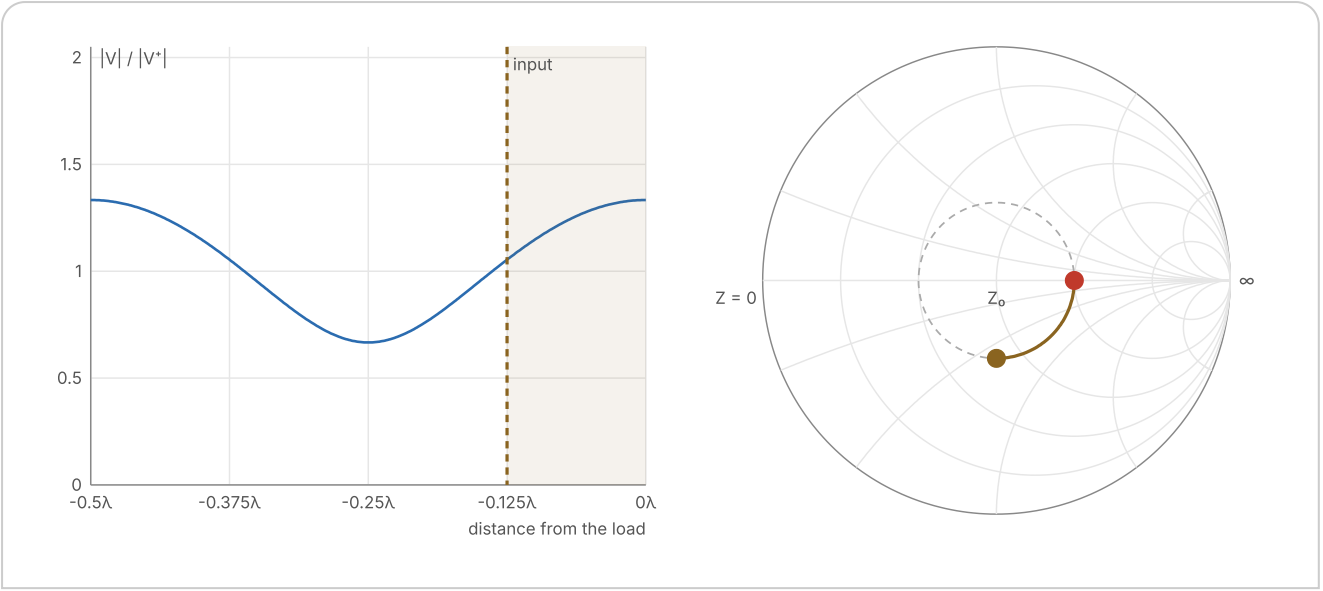
Without losses they give a wave with speed $1/\sqrt{L'C'}$ and characteristic impedance $Z_0 = \sqrt{L'/C'}$ – 50 Ω or 75 Ω for standard cables. It is the same mathematics as for sound in a tube, where pressure corresponds to voltage, volume velocity to current, the mass of the air to L' and its compressibility to C'. Heaviside also realised that distortion can be avoided if $R'/L' = G'/C'$, and proposed "loading" telephone lines with extra coils – an idea that made long-distance telephony possible.

Reflection and the Smith chart

If the line ends in a load $Z_L \neq Z_0$, part of the wave is reflected with the reflection coefficient $\Gamma_L = (Z_L - Z_0)/(Z_L + Z_0)$, and a standing-wave pattern arises with the standing-wave ratio $\mathrm{SWR} = (1 + |\Gamma|)/(1 - |\Gamma|)$. Seen through a line of length l the impedance is transformed:

$$Z_{\text{in}} = Z_0 \,\frac{Z_L + jZ_0 \tan \beta l}{Z_0 + jZ_L \tan \beta l}, \qquad \beta = \frac{2\pi}{\lambda}$$

A quarter-wave line turns a short circuit into an open circuit and vice versa, and transforms Z_L into Z_0^2/Z_L – the quarter-wave transformer. In 1939 Phillip Smith of Bell Labs drew a diagram in which the whole complex impedance plane is mapped into the unit circle for Γ. On the Smith chart one moves along the line simply by rotating about the centre. It is still the radio engineer's most important graphical tool – and it is on the screen of every network analyser.



Waveguides

At microwave frequencies hollow metal tubes are used as waveguides. Inside the tube the fields can exist only in certain patterns, modes, and each mode has a cut-off frequency. In a rectangular waveguide of width a the lowest mode, TE_{10}, has cut-off at $f_c = c/2a$. Below cut-off the wave dies away exponentially – which is why microwaves do not escape through the holes in an oven door. Above cut-off the propagation constant is

$$\beta = \frac{\omega}{c}\sqrt{1 - \left(\frac{f_c}{f}\right)^2}$$

Exactly the same holds for sound in a hard duct: the plane wave always propagates, but the transverse modes have a cut-on frequency at which half a transverse wavelength fits into the duct. In a lined duct the propagation constants become complex, and the modes can behave surprisingly – that was the core of my thesis.

Antennas

An antenna is the transition from a wave bound to a line to a free wave. Heinrich Hertz demonstrated electromagnetic waves in 1887 with a dipole. A short dipole of length $l \ll \lambda$ radiates the power $P = \tfrac{1}{2} R_{\mathrm{rad}} I_0^2$ with the radiation resistance $R_{\mathrm{rad}} = 80\pi^2 (l/\lambda)^2\ \Omega$. A half-wave dipole has $R_{\mathrm{rad}} \approx 73\ \Omega$, close to the impedance of a coaxial cable. Antennas are reciprocal: the radiation

pattern is the same for transmission and reception – the same principle as for microphones and loudspeakers (see electroacoustics).

Since 1989

- **Computational electromagnetics:** FDTD (Yee, 1966) and finite element programs such as COMSOL and HFSS now compute the fields around a whole mobile phone or aircraft.
- **Signal integrity:** With clock frequencies in the GHz range every circuit-board track is a transmission line; impedance matching and reflections are everyday problems for digital designers.
- **Mobile and Wi-Fi antennas** have become small printed antennas and antenna arrays with electronically steered beams (5G MIMO).
- **Metamaterials** with negative refractive index (Pendry, Smith, around 2000) and transformation optics use the coordinate invariance of Maxwell's equations to design "invisibility cloaks" – and acoustic versions of the same.

Further reading

- David K. Cheng: *Field and Wave Electromagnetics* (2nd ed., Addison-Wesley 1989)
- Simon Ramo, John R. Whinnery and Theodore Van Duzer: *Fields and Waves in Communication Electronics* (3rd ed., Wiley 1994)
- David M. Pozar: *Microwave Engineering* (4th ed., Wiley 2012)
- On this site: James Clerk Maxwell · The electromagnetic spectrum